

Ivan Valentini

Pentester | Vulnerability Researcher

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Experience

Cybersecurity Engineer, H Company – Paris, France

Nov 2024 – present

Worked on [Fluorite](#), a Rust framework for running distributed enclave applications on Kubernetes. It is designed for deploying attested, unalterable workloads in trusted and confidential computing environments. I also lead internal penetration tests and vulnerability assessments for company products, platforms, and AI systems.

- Conducted internal penetration tests and vulnerability assessments for our products, platforms, and AI systems.
- Lead the deployment of secure AI agent sandboxes in production.
- Secured AWS cloud environment by identifying vulnerabilities, hardening configurations, and implementing security best practices.

Vulnerability Researcher Intern, Quarkslab – Paris, France

Mar 2024 – Sept 2024

Focused on researching and analyzing security vulnerabilities in [gVisor](#), a secure container runtime developed by Google.

- Researched gVisor's architecture, analyzing how its user-space application kernel implements Linux system calls to isolate containers and secure the host.
- Performed static and dynamic code analysis and used debugging tools to conduct comprehensive security assessments and threat modeling.
- Examined gVisor's sandboxing mechanisms, identifying potential attack vectors and vulnerabilities that could compromise container security.

Python Developer Intern, Fondazione Bruno Kessler – Trento, Italy

Feb 2022 – Apr 2022

Focused on learning about recently discovered TLS vulnerabilities and implementing checks for them in [TLSAssistant](#), a tool developed at FBK for assessing the security of TLS servers.

- Developed Python scripts to automate the analysis of TLS server configurations and identify potential security vulnerabilities.
- Implemented checks for recently discovered TLS vulnerabilities, including ALPACA, POODLE, and RACCOON, to enhance the capabilities of TLSAssistant, a tool developed at FBK for assessing server security.
- Contributed to the development and improvement of open-source TLS scanners, such as [TLS-Scanner](#), by implementing new features and enhancing existing functionalities.

Education

EURECOM / University of Trento, Master's degree in Cybersecurity – Trento, Italy

Sept 2022 – July 2024

Double joint degree graduated cum laude in Cybersecurity from EURECOM and the University of Trento, offered by EIT Digital, specializing in mobile device and cloud security.

University of Trento, Bachelor's degree in Computer Science – Trento, Italy

Sept 2019 – July 2022

Three-year bachelor's degree from the University of Trento, covering algorithms, data structures, databases, and the fundamentals of statistics and machine learning.

Projects

[Fluorite](#) – H Company, Paris, France

Nov 2024 – Feb 2026

- Built Fluorite, a Rust framework for deploying attested, immutable workloads in trusted and confidential computing environments on Kubernetes with support for Microsoft Azure, Google Cloud, and AMD SEV-SNP bare-metal, alongside NVIDIA H100 GPUs in confidential mode
- Implemented end-user-verifiable cluster-wide attestation and Attested TLS to bind certificates to hardware-backed integrity proofs.
- Demonstrated the framework by deploying a secure, distributed, and attested AI chat application.

[Fitbit wearable security research](#) – EURECOM, France

Oct 2023 – Feb 2024

Conducted security research on the Golden Gate communication framework used between Fitbit wearables and Fitbit app over Bluetooth Low Energy (BLE).

- Analyzed network architecture and CoAP messaging behavior between the Fitbit Charge 5 and the official Fitbit companion app.
- Built custom JavaScript hooking scripts with Frida to track low-level traffic and extract clear-text communication across the framework.

[CyberChallenge.IT](#) – Italy

Sept 2022 – June 2023

Training program organized by the Italian Cybersecurity National Lab.

- Selected among the students of my university to compete in nationwide high-stakes Attack and Defense CTF competition.
- Acquired deep knowledge across core security domains, including web security, reverse engineering, binary analysis, software vulnerabilities, cryptography, and hardware security.
- Developed custom tools and exploit scripts using Python, C, C++, and Rust to automate exploitation of targets.

Awards

- [Won](#) the first edition of the CTF competition hosted by [Würth Phoenix](#), and got [second place](#) on the second edition.
- 15th place on [Websec.fr](#), a CTF platform focused on PHP challenges (Oct. 2023)
- 81st place on [Reyammer CTF](#), a CTF platform focused on Android challenges (Jul. 2023)
- More than 70 systems owned on [HackTheBox](#). Hundreds of security challenges completed and flags captured on [TryHackMe](#) and [RootMe](#)

Skills

Programming Languages: Python, Rust, C, Bash.

Pentesting Tools: Burp Suite, Frida, Ghidra.

Infrastructure: Kubernetes, Docker, Azure, AWS, Linux.

Interest Areas: Vulnerability Research, Penetration Testing, Reverse Engineering, Trusted Execution Environments, Confidential Computing, Secure Software Development.

Languages: English (C2), French (B2), Italian (Native).